

Towards Real-time Privacy-preserving Audio-Visual Speech Enhancement

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Abstract

Human auditory cortex in everyday noisy situations is known to exploit aural and visual cues that are contextually combined by the brain's multi-level integration strategies to selectively suppress the background noise and focus on the target speaker. The multimodal nature of speech is well established, with listeners known to unconsciously lip read to improve the intelligibility of speech in noise. However, despite significant research in the area of audio-visual (AV) speech enhancement real-time processing models, with low latency remains a formidable technical challenge. In this paper, we propose a novel audio-visual speech enhancement model based on Temporal Convolutional Networks (TCN) that exploit the privacy preserving lip-landmark flow features for speech enhancement in multi-talker cocktail party environments. In addition, we propose an efficient implementation of TCN, called Fast-TCN, to enable real time deployment of the proposed framework. The comparative simulation results in terms of speech quality and intelligibility demonstrate the effectiveness of our proposed AV model as compared to benchmark audio-only and audio-visual approaches for speaker and noise independent scenarios.

Index Terms: speech enhancement, audio-visual speech separation, privacy-preserving

1. Introduction

Speech enhancement (SE) aims to improve the overall intelligibility and quality of the speech degraded with background interference. SE finds application in a wide range of real world problems including hearing aids, smart human-computer interaction systems, teleconferencing, and automatic speech recognition [1]. Conventional single-channel SE methods including spectral subtraction and Wiener Filtering perform well in the presence of stationary noise but fail in environments where non-stationary noise is present. More recently, the application of supervised learning algorithms have shown significant improvement in generalisation performance compared to traditional SE algorithms [2, 3].

Despite significant research in the area of SE, understanding speech in the presence of multiple competing background sources, that are often encountered in cocktail party situations, has been a key challenge for decades [4]. Therefore, researchers have proposed audio-visual (AV) speech enhancement models [5, 6, 7, 8, 9, 10, 11] that have shown to achieve significant performance improvement as compared to audio-only SE models. For example, Morrone et al. [5] introduced a model to address the problem of enhancing speech in cocktail party scenarios using facial landmarks. The AV features are fed to a bidirectional LSTM model to predict spectral time frequency masks for SE. Experimental results reveal the generalisation capability of the model trained using TCD-TIMIT and Grid corpus in speaker independent settings. In addition, Wang et al. [6] proposed a AV SE framework that exploits the correlation be-

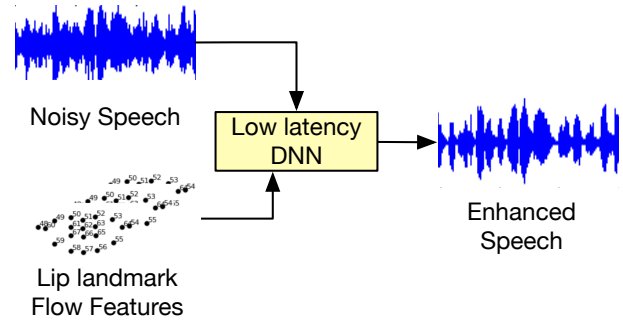


Figure 1: Schematic diagram of proposed framework

tween visual lip images and noisy speech for more effective SE. More recently, Ramesh et al. [12] proposed a SE model based on transformers to improve the AV SE performance for learning long-term dependency for both audio and visual features.

However, despite significant research in the area of AV SE limited work has been conducted to develop AV models that can be deployed in real-time applications including hearing aids and cochlear implants as the AV SE models are non-causal and computationally complex. In addition, the use of images in most state-of-the-art AV SE framework introduces privacy concerns for cloud deployment of such models in real-world applications.

In this work, we propose a novel real-time AV SE framework that exploits privacy preserving lip landmark flow features to isolate a speaker's voice from interfering background speech and noise sources. The framework outputs an enhanced audio signal containing only the target speaker when a noisy audio mixture consisting of target speaker mixed with multiple competing noise sources and landmark flow features are fed as input. It is to be noted that, the speaker face cannot be reconstructed from landmark flow features.

An overview of the proposed framework is depicted in Figure 1. The proposed model exploits the privacy preserving lip landmark flow features and noisy speech for optimal SE in real-world unconstrained environments. Specifically, a unified deep neural network model based on temporal convolutional network effectively learns correlations between noisy speech and lip optical flow features to produce a spectral ratio mask. The spectral mask is multiplied with noisy spectrogram to preserve the target speech dominant regions and suppress background noise dominant regions. The enhanced speech is resynthesised by combining the masked spectrogram with noisy phase. To the best of our knowledge, our paper is the first to propose a privacy-preserving real-time AV SE framework that is speaker and noise independent.

The rest of the paper is organised as follows: Section 2 presents synthetic AV corpora and data preprocessing, Section 3 presents the proposed lip landmark real-time AV SE framework, section 4 presents the experimental results and finally section 5

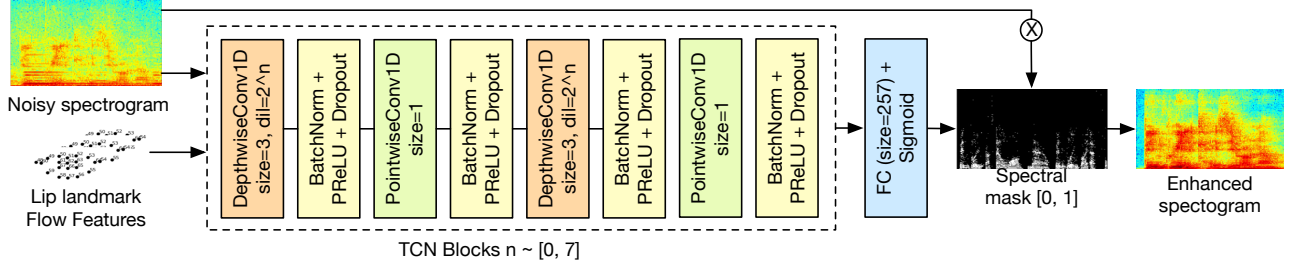


Figure 2: Model architecture

concludes the work and propose future directions.

2. Audio-Visual Dataset and Preprocessing

2.1. Dataset

In this section, we discuss the AV corpora used to evaluate the performance of the proposed approach. The models are trained and evaluated using widely used GRID [13] and TCD-TIMIT [14] corpora. Specifically, two different test scenarios were considered: (1) two speaker mixture - target speaker is mixed with a background speaker at randomly selected SNR ranging from a uniform distribution between 0 dB and +10 dB (2) Two speaker with real ambient background noise mixture - target speaker mixed with background speaker (at SNR ranging from 0 dB to +10 dB) and background noise (at SNR ranging from -6 dB to +6 dB). It is to be noted that, the WHAM corpus [15] is used as ambient noise in generating two speaker with real ambient background noise mixture. WHAM consists of ambient noises recorded in coffee shops, restaurants, and bars in the San Francisco Bay Area.

2.1.1. GRID

: In our experiment, we used a widely benchmark Grid dataset which consists of 33 speakers along with 1000 utterances for each speaker. The sentence structure is based on six words [13]. For training, validation and testing, the data was split into 25, 4, and 4 speakers respectively.

2.1.2. TCD-TIMIT

: Furthermore to generalise our model, we used TCD-TIMIT corpus which consists of 5488 utterances and 56 speakers [14]. For training, validation and testing, the data was split into 45, 6, and 6 speakers respectively.

2.2. Preprocessing

2.2.1. Audio

The audio signal is resampled to 16000 Hz sampling frequency and then segmented into 32 ms frames with a 25% frame increment. The short time Fourier transform with hanning window is used to generate a 257 bin spectrum. It is to be noted that, the magnitude of the spectrum is fed as input to the model.

2.2.2. Video

BlazeFace [16] is used to extract the face region from the videos resampled at 25 fps. The region is resized to 224×224 . The cropped region is then fed to Face Mesh [17] model to extract 486 dimensional landmarks per frame. There is a high correla-

tion between the audio and lip movements, only the lip part of the Face Mesh feature is considered resulting 40 dimensional 3D lip landmark features per frame. The landmark of each current frame is subtracted from the previous frame to generate an optical flow of landmarks as visual features. The visual features are upsampled to match the audio feature sampling rate.

3. Real-time Privacy-preserving Audio-Visual Speech Enhancement

3.1. Model Architecture

This section describes the network architecture depicted in Fig. 2. The network consists of Temporal convolutional network (TCN) blocks and a fully connected layer. Each individual TCN block consists of two depth-wise 1D convolutional layer (size = 3) followed by batch norm, PReLU activation, dropout (0.1) and point-wise 1D convolutional layer (size = 1) as shown in Fig. 2. The proposed model consists of 8 TCN blocks where dilation of the convolutional layers is set to 2^n where n ranges from 0 to 7.

The fully connected layers consists of 257 neurons with sigmoid activation. The predicted mask from fully connected layer is multiplied with noisy spectrogram to output enhanced speech spectrogram. The model is trained to minimise the mean absolute error between masked spectrogram and clean speech spectrogram. The predicted masked spectrogram is combined with noisy phase spectrum and a inverse short time Fourier transform is applied to resynthesise enhanced speech.

3.2. Optimisation for Real-time Prediction

The prediction time for the naive implementation of the proposed model increases exponentially ($O(2^L)$) with the number of TCN blocks due to redundant convolution operations. This limits the usability of model in real-time applications. We address this limitation by caching the outputs of intermediate layer by replacing zero padding layers with contextual padding layers. This speeds up the model prediction time from $O(2^L)$ to $O(L)$ as show in Fig. 3.

4. Results

4.1. Experimental Setup

The proposed framework is developed in Pytorch and trained using Titan Xp GPUs. The speakers present in GRID and TCD-TIMIT corpora are split into train (70%), validation (15%) and test sets (15%) for speaker independent evaluation. It is to be noted that, the speakers are divided to ensure similar gender representation across all sets. The ambient noises data is also separated into train, validation and test set to ensure noise in-

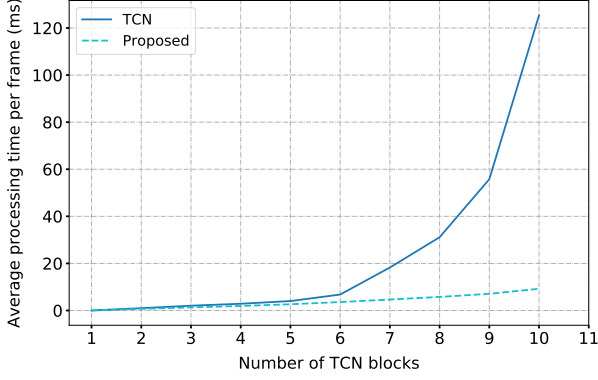


Figure 3: Comparison of TCN vs. Fast TCN

Table 1: Objective evaluation: GRID 2 speaker mixture

	PESQ	STOI	CSIG	CBAK	COVL
Noisy	2.063	0.626	2.625	1.909	1.938
ConvTasNet [18]	2.206	0.631	2.733	1.993	2.051
LandmarkSE [5]	2.511	0.646	3.083	2.095	2.464
Proposed AV	3.041	0.685	3.350	2.141	2.638
Oracle IRM	3.038	0.681	3.188	1.907	2.495
Oracle PSM	3.168	0.710	3.829	2.378	3.134

dependent evaluation. The model is trained with an Adam optimiser ($\text{lr}=1\text{e-}3$) for 50 epochs. The learning rate is multiplied by 0.9 when the model validation accuracy stops decreasing for 2 consecutive epochs. The model with best validation accuracy is used for evaluation.

4.2. Objective testing on Synthetic mixtures

The quality and intelligibility of SE is evaluated using five widely used evaluation metrics including (1) PESQ: the degraded signal is compared with reference signal and it usually provides score from 0.5 to 4.5 [19] (2) short-time objective Intelligibility (STOI) [20] provides score from 0 to 1 (3) CSIG: mean opinion score to estimate the signal distortion and ranges from 1 to 5 [21] (4) CBAK: used to predict the distortion due to background noise and ranges from 1 to 5 [21] (5) COVL: used to estimate the overall quality and ranges from 1 to 5 [21].

The proposed AV model is compared with ConvTasNet [18], LandmarkSE [5] and spectral mask targets including ideal ratio mask (IRM) and phase sensitive mask (PSM). The objective comparison result of ConvTasNet, our proposed AV, LandmarkSE, Oracle IRM, Oracle PSM, for Grid and TCD-TIMIT corpus are presented in Table 1, 2, 3 and 4. It can be seen that for all objective measures the proposed real-time privacy-preserving audio-visual speech enhancement model outperforms ConvTasNet, and LandmarkSE for both AV corpora. In addition, the AV model achieves performance similar to oracle IRM.

Figure 4 depicts a spectrogram of resynthesised speech generated using ConvTasNet and proposed AV model for a randomly selected utterances from TCD-TIMIT 2 Speakers + Noise mixture test set. The enhanced spectrogram is compared with both reference (clean) speech, oracle PSM and degraded (noisy) speech. It can be seen that our proposed model achieve near ideal spectrogram reconstruction as compared to ConvTas-

Table 2: Objective evaluation: GRID 2 speaker + WHAM! Noise mixture

	PESQ	STOI	CSIG	CBAK	COVL
Noisy	1.610	0.575	1.998	1.630	1.426
ConvTasNet [18]	1.806	0.581	2.233	1.793	1.622
LandmarkSE [5]	2.122	0.629	2.483	1.891	2.132
Proposed AV	2.745	0.676	3.224	2.038	2.493
Oracle IRM	2.528	0.678	2.461	1.719	1.987
Oracle PSM	2.944	0.703	3.437	2.099	2.727

Table 3: Objective evaluation: TCD-TIMIT 2 speaker mixture

	PESQ	STOI	CSIG	CBAK	COVL
Noisy	1.927	0.588	1.974	1.818	1.588
ConvTasNet [18]	2.106	0.597	2.221	1.833	1.751
LandmarkSE [5]	2.412	0.616	2.483	1.899	2.012
Proposed AV	2.911	0.671	2.938	1.921	2.481
Oracle IRM	2.921	0.672	3.062	1.923	2.546
Oracle PSM	3.399	0.694	3.646	2.328	3.095

net.

4.3. Processing latency

The processing latency for a real-time AV SE algorithm is defined as the difference between the time arrival of speech and the time when the algorithm finish processing. The speech data is processed frame-by-frame and for real-time processing the algorithm should finish the processing each width of frame within the shift time. The frame width and shift time for the proposed framework are 32ms and 8ms respectively. The conversion of time domain frame to frequency domain and vice-versa requires 1ms. The proposed model prediction time for a single frame is 3ms. It can be seen that, the total time taken by preprocessing and the model prediction is less than the frame shift time. Therefore, the proposed model can be used in real-time applications. It is to be noted that, these values are computed using 2.2 GHz Quad-Core Intel Core i7 with 16 GB RAM.

5. Conclusion

This paper presented a real-time privacy preserving audio-visual (AV) speech enhancement model based on Temporal Convolutional Networks (TCN) that exploits privacy preserving landmark flow features. We tested proposed framework on benchmark GRID and TCD-TIMIT data and achieved superior results as compared to state-of-the-art speech enhancement models including ConvTasNet and LandmarkSE. In addition, we presented an optimised implementation of the proposed model (Fast-TCN) to enable real-time processing on consumer level CPU. Ongoing work includes comparison with other state-of-the-art approaches using subjective listening tests. In future, we intend to evaluate the proposed framework with more challenging AV speech corpora.

6. Acknowledgements

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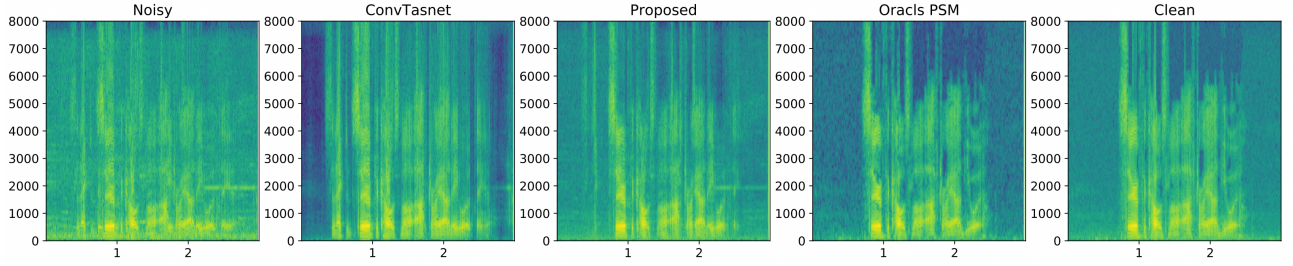


Figure 4: Spectrogram comparison for TCD TIMIT: 2 Speakers + Noise mixture

Table 4: Objective evaluation: TCD-TIMIT 2 speaker + WHAM! Noise mixture

	PESQ	STOI	CSIG	CBAK	COVL
Noisy	1.930	0.568	1.708	1.633	1.378
ConvTasNet [18]	2.106	0.571	1.933	1.693	1.451
LandmarkSE [5]	2.406	0.596	2.483	1.725	1.764
Proposed AV	2.692	0.628	2.638	1.855	2.115
Oracle IRM	2.719	0.666	2.587	1.722	2.164
Oracle PSM	3.197	0.693	3.356	2.087	2.782

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